

Convex Hull

1 Problem Definition

Convex Hull

Given a finite set of planar points

$$S = \{P_1, P_2, \dots, P_n\},$$

the **convex hull** of S is the smallest convex set that contains every point of S . When the points are not all collinear, this set is a convex polygon.

The input is a set of points:

$$S = \{P_1, P_2, \dots, P_n\}.$$

The output is an ordered list of boundary vertices:

$$H = (H_1, H_2, \dots, H_m), \quad m \leq n.$$

The vertices $H_1, \dots, H_m$ are the outermost points. Points strictly inside the polygon are not hull vertices.

Rubber-Band Interpretation

Imagine stretching a rubber band around all the points and then releasing it. The rubber band touches the outer boundary points. Those touched points form the convex hull.

2 Why It Is Called “Convex”

A set is convex if the line segment between any two points in the set stays inside the set. The convex hull is the smallest convex shape that contains the original point set.

Smallest Convex Container

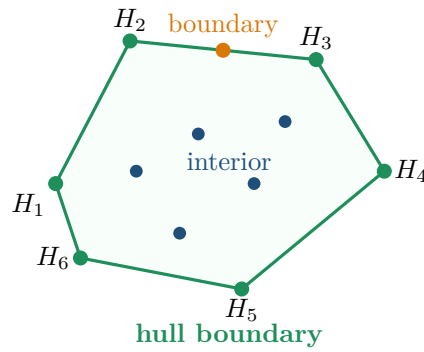
$$\text{CH}(S) = \text{the smallest convex region containing } S.$$

For algorithm problems, we usually return only the boundary vertices of this region, not every point inside the region.

3 Hull Vertices, Boundary Points, and Interior Points

For a point set, a point may be:

Type	Meaning
Hull vertex	A corner point of the outer boundary.
Boundary point	A point on an edge of the hull but not necessarily a corner.
Interior point	A point strictly inside the hull; not part of the hull vertex list.



Boundary Convention

Some algorithms return every point on the boundary, including collinear edge points. Many convex-hull algorithms return only the corner vertices. In these notes, “hull vertices” means the corner points unless stated otherwise.

4 Convex Polygon

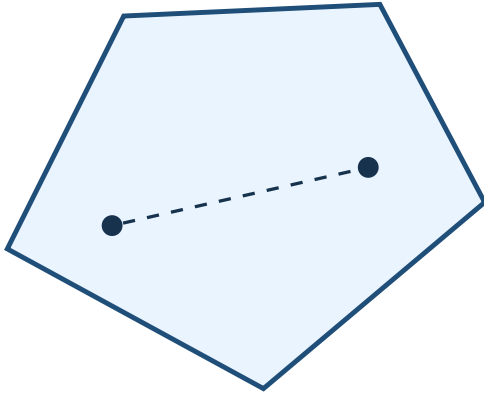
Convex Polygon

A polygon is **convex** if it does not bend inward.

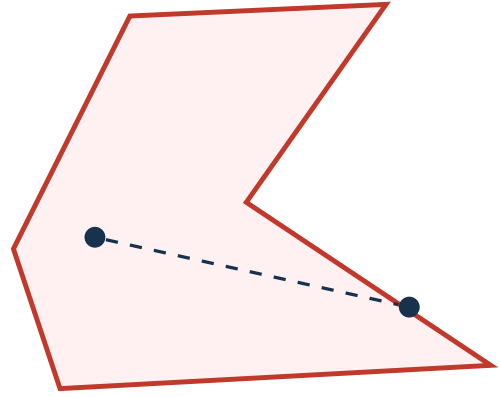
Equivalent tests for convexity:

- every interior angle is at most 180° ;
- for a strictly convex polygon, every interior angle is less than 180° ;
- every line segment joining two points inside the polygon stays inside the polygon;
- when the boundary is traversed in one direction, there are no opposite-sign turns; for a strictly convex polygon, all turns have the same nonzero orientation.

5 Convex vs Non-Convex



convex polygon
segment stays inside



non-convex polygon
segment leaves polygon

6 Orientation View of the Hull

Convex hull algorithms depend heavily on orientation tests. For three points

$$A = (A_x, A_y), \quad B = (B_x, B_y), \quad C = (C_x, C_y),$$

the orientation value is

$$\text{orient}(A, B, C) = (B_x - A_x)(C_y - A_y) - (B_y - A_y)(C_x - A_x).$$

Orientation Test

$$\begin{aligned} \text{orient}(A, B, C) > 0 &\implies \text{left turn,} \\ \text{orient}(A, B, C) < 0 &\implies \text{right turn,} \\ \text{orient}(A, B, C) = 0 &\implies \text{collinear points.} \end{aligned}$$

If the hull vertices are listed counterclockwise as

$$H_1, H_2, \dots, H_m,$$

then every consecutive boundary turn should be a left turn:

$$\text{orient}(H_i, H_{i+1}, H_{i+2}) > 0,$$

where indices wrap around the boundary.

Same-Turn Rule

A convex polygon has no inward turn when traversed around its boundary.

For a clockwise hull, all nonzero turns should be right turns. If a turn has the opposite sign from the traversal direction, the polygon bends inward at that point.

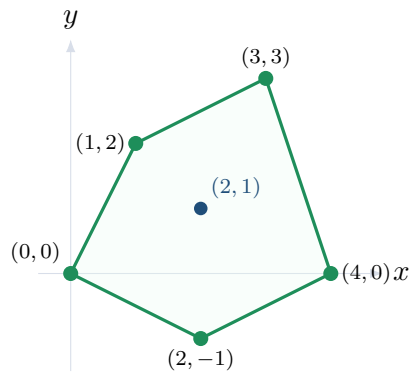
7 Example 1: One Interior Point

Problem

Given

$$S = \{(0, 0), (1, 2), (2, 1), (3, 3), (4, 0), (2, -1)\},$$

identify the hull vertices.



The point $(2, 1)$ lies inside the polygon, so it is not a hull vertex.

Answer

One clockwise ordering of the convex hull is

$$(0, 0), (1, 2), (3, 3), (4, 0), (2, -1).$$

8 Example 2: All Points Collinear

Problem

Find the convex hull of

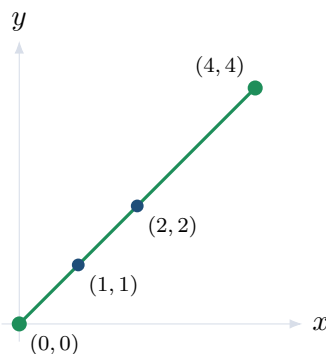
$$S = \{(0, 0), (1, 1), (2, 2), (4, 4)\}.$$

All points lie on the same line:

$$y = x.$$

For example,

$$\text{orient}((0, 0), (1, 1), (4, 4)) = 1 \cdot 4 - 1 \cdot 4 = 0.$$



Answer

If only extreme hull vertices are required, the hull is the line segment

$$(0, 0) \rightarrow (4, 4).$$

The middle points are boundary points but not corner vertices.

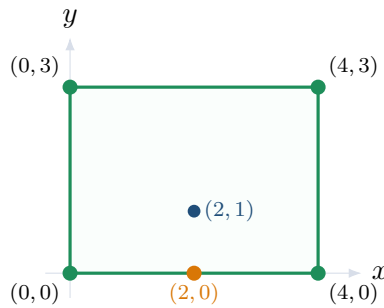
9 Example 3: Boundary Collinear Point

Problem

Consider

$$S = \{(0, 0), (2, 0), (4, 0), (4, 3), (0, 3), (2, 1)\}.$$

Which points are hull vertices?



The point $(2, 0)$ lies on the bottom edge. It is on the boundary, but it is not a corner. The point $(2, 1)$ is strictly inside.

Answer

The hull vertices are

$$(0, 0), (4, 0), (4, 3), (0, 3).$$

10 Orientation Check on a Hull

Use the hull from Example 1:

$$H_1 = (0, 0), \quad H_2 = (1, 2), \quad H_3 = (3, 3).$$

Check the first turn:

$$\overrightarrow{H_1 H_2} = (1, 2), \quad \overrightarrow{H_1 H_3} = (3, 3).$$

$$\text{orient}(H_1, H_2, H_3) = 1 \cdot 3 - 2 \cdot 3 = 3 - 6 = -3.$$

This value is negative because the listed order in Example 1 is clockwise, not counterclockwise. The same-turn rule still applies: a clockwise hull should have consistently negative turns.

Ordering Matters

The hull may be listed clockwise or counterclockwise. What matters is consistency: all boundary turns should have the same sign for a convex polygon.

11 Final Review

Convex Hull Summary

The convex hull is the smallest convex set containing all points.
Hull vertices are outermost corner points.
Interior points are excluded from the hull vertex list.
Boundary collinear points may be excluded depending on convention.
A valid convex boundary has consistent orientation signs.